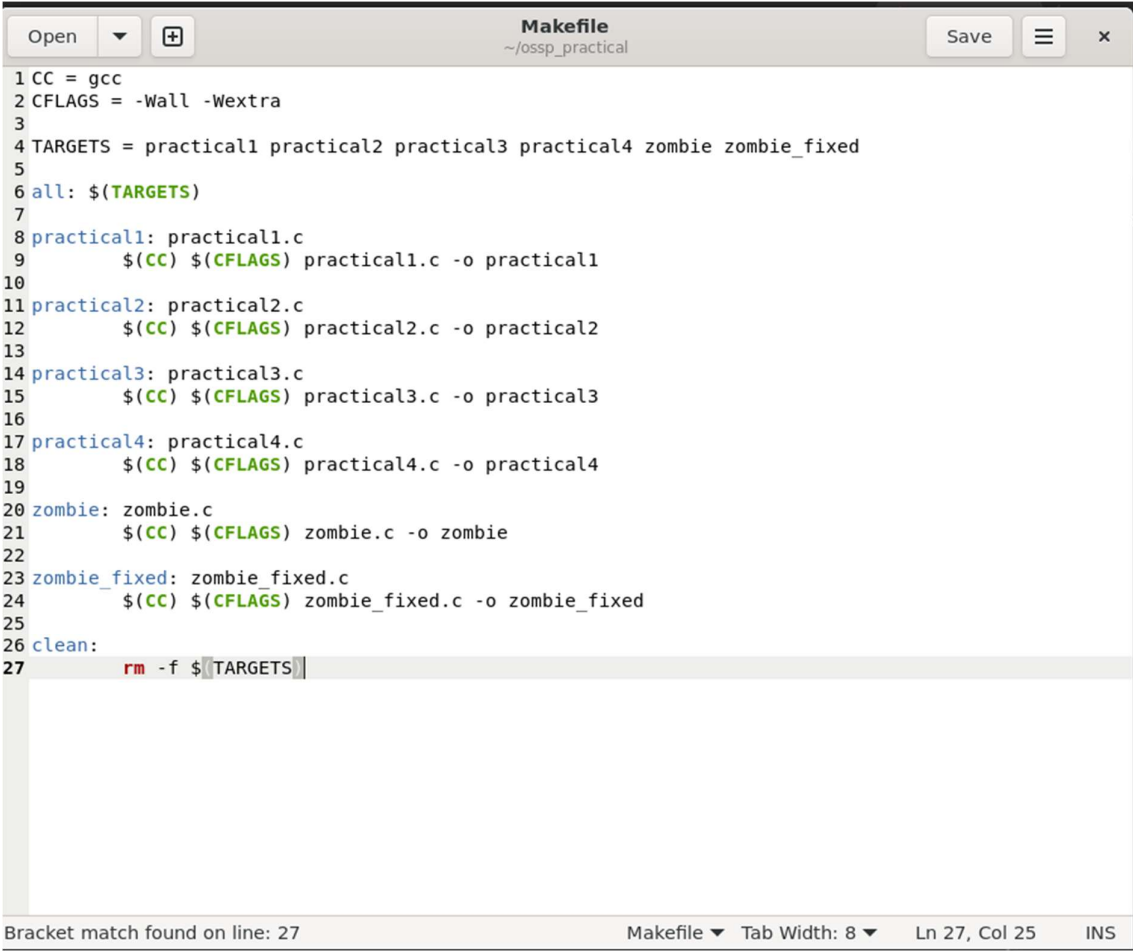


# Skill -1 OS



```
1 CC = gcc
2 CFLAGS = -Wall -Wextra
3
4 TARGETS = practical1 practical2 practical3 practical4 zombie zombie_fixed
5
6 all: $(TARGETS)
7
8 practical1: practical1.c
9     $(CC) $(CFLAGS) practical1.c -o practical1
10
11 practical2: practical2.c
12     $(CC) $(CFLAGS) practical2.c -o practical2
13
14 practical3: practical3.c
15     $(CC) $(CFLAGS) practical3.c -o practical3
16
17 practical4: practical4.c
18     $(CC) $(CFLAGS) practical4.c -o practical4
19
20 zombie: zombie.c
21     $(CC) $(CFLAGS) zombie.c -o zombie
22
23 zombie_fixed: zombie_fixed.c
24     $(CC) $(CFLAGS) zombie_fixed.c -o zombie_fixed
25
26 clean:
27     rm -f $(TARGETS)
```

Bracket match found on line: 27      Makefile    Tab Width: 8    Ln 27, Col 25    INS

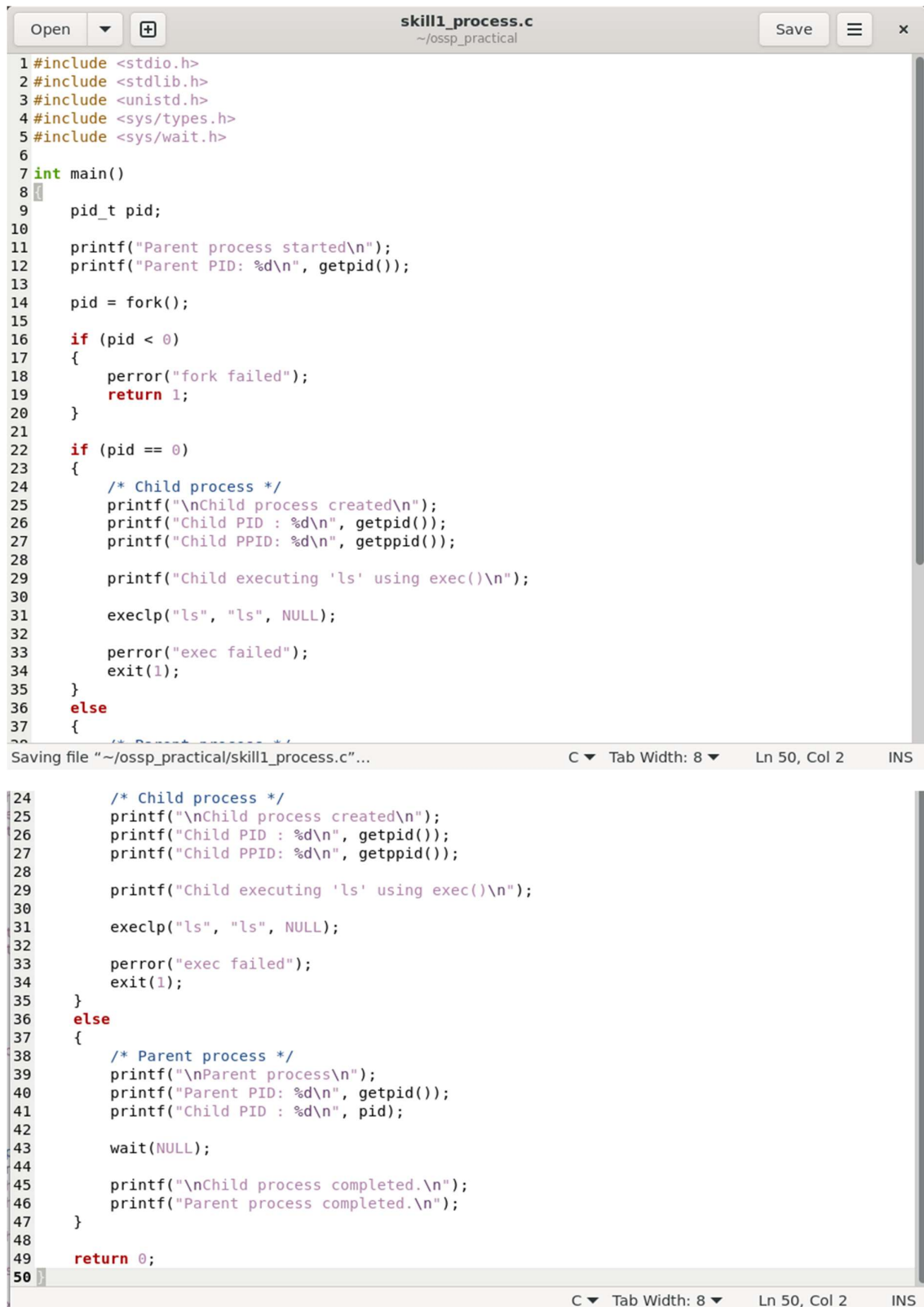
# Skill -1 OS

```
siddharth@SiddharthReddy:~$ cd ~/ossp_practical
siddharth@SiddharthReddy:~/ossp_practical$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

siddharth@SiddharthReddy:~/ossp_practical$ git --version
git version 2.43.0
siddharth@SiddharthReddy:~/ossp_practical$ ls -la
total 156
drwxr-xr-x 2 siddharth siddharth 4096 Aug 12 09:03 .
drwxr-x--- 8 siddharth siddharth 4096 Aug 12 06:54 ..
-rw-r--r-- 1 siddharth siddharth 34 Aug 12 07:03 destination.txt
-rwxr-xr-x 1 siddharth siddharth 16480 Aug 12 06:56 practical1
-rw-r--r-- 1 siddharth siddharth 1014 Aug 12 06:55 practical1.c
-rwxr-xr-x 1 siddharth siddharth 16224 Aug 12 07:03 practical2
-rw-r--r-- 1 siddharth siddharth 1185 Aug 12 07:02 practical2.c
-rwxr-xr-x 1 siddharth siddharth 16264 Aug 12 08:28 practical3
-rw-r--r-- 1 siddharth siddharth 1054 Aug 12 08:28 practical3.c
-rwxr-xr-x 1 siddharth siddharth 16400 Aug 12 08:55 practical4
-rw-r--r-- 1 siddharth siddharth 2139 Aug 12 08:55 practical4.c
-rw-r--r-- 1 siddharth siddharth 11 Aug 12 07:08 sample.txt
-rw-r--r-- 1 siddharth siddharth 34 Aug 12 07:03 source.txt
-rw-r--r-- 1 siddharth siddharth 8102 Aug 12 07:10 trace.txt
-rwxr-xr-x 1 siddharth siddharth 16216 Aug 12 08:58 zombie
-rw-r--r-- 1 siddharth siddharth 736 Aug 12 08:58 zombie.c
-rwxr-xr-x 1 siddharth siddharth 16272 Aug 12 09:03 zombie_fixed
-rw-r--r-- 1 siddharth siddharth 953 Aug 12 09:03 zombie_fixed.c
siddharth@SiddharthReddy:~/ossp_practical$ gedit Makefile
```

```
siddharth@SiddharthReddy:~/ossp_practical$ gedit Makefile
siddharth@SiddharthReddy:~/ossp_practical$ make
make: Nothing to be done for 'all'.
siddharth@SiddharthReddy:~/ossp_practical$ make -n
make: Nothing to be done for 'all'.
siddharth@SiddharthReddy:~/ossp_practical$ make -B
gcc -Wall -Wextra practical1.c -o practical1
gcc -Wall -Wextra practical2.c -o practical2
gcc -Wall -Wextra practical3.c -o practical3
gcc -Wall -Wextra practical4.c -o practical4
gcc -Wall -Wextra zombie.c -o zombie
gcc -Wall -Wextra zombie_fixed.c -o zombie_fixed
siddharth@SiddharthReddy:~/ossp_practical$ ls -l Makefile
-rw-r--r-- 1 siddharth siddharth 562 Aug 13 08:14 Makefile
siddharth@SiddharthReddy:~/ossp_practical$ |
```

# Skill -1 OS



```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main()
8 {
9     pid_t pid;
10
11     printf("Parent process started\n");
12     printf("Parent PID: %d\n", getpid());
13
14     pid = fork();
15
16     if (pid < 0)
17     {
18         perror("fork failed");
19         return 1;
20     }
21
22     if (pid == 0)
23     {
24         /* Child process */
25         printf("\nChild process created\n");
26         printf("Child PID : %d\n", getpid());
27         printf("Child PPID: %d\n", getppid());
28
29         printf("Child executing 'ls' using exec()\n");
30
31         execlp("ls", "ls", NULL);
32
33         perror("exec failed");
34         exit(1);
35     }
36     else
37     {
38         /* Parent process */
39         printf("\nParent process\n");
40         printf("Parent PID: %d\n", getpid());
41         printf("Child PID : %d\n", pid);
42
43         wait(NULL);
44
45         printf("\nChild process completed.\n");
46         printf("Parent process completed.\n");
47     }
48
49     return 0;
50 }
```

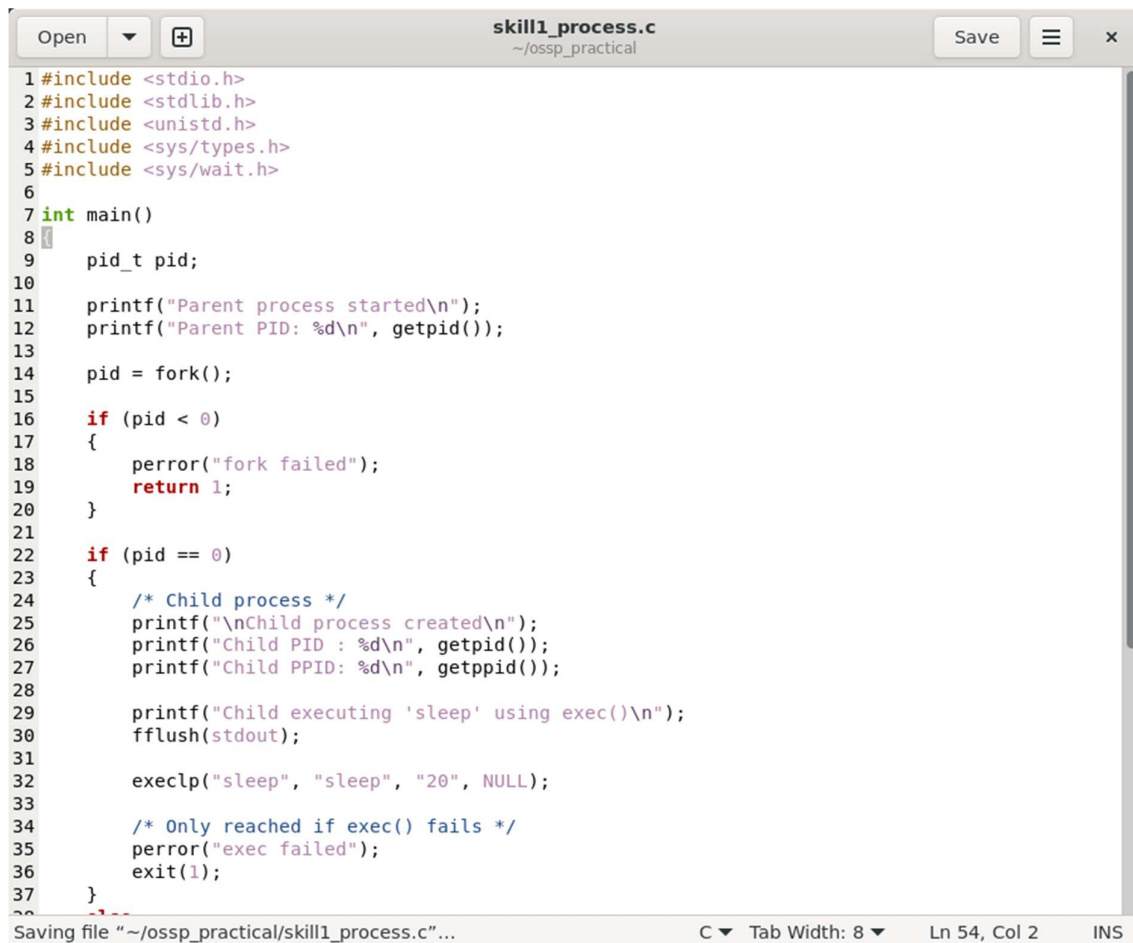
# Skill -1 OS

```
siddharth@siddharthReddy:~/ossp_practical$ ./skill1_process
Parent process started
Parent PID: 16068

Parent process
Parent PID: 16068
Child PID : 16069

Child process created
Child PID : 16069
Child PPID: 16068
Child executing 'ls' using exec()
Makefile      practical1.c  practical3    practical4.c  skill1_process.c  zombie        zombie_fixed.c
destination.txt practical2    practical3.c  sample.txt    source.txt        zombie.c
practical1     practical2.c practical4     skill1_process trace.txt         zombie_fixed

Child process completed.
Parent process completed.
siddharth@siddharthReddy:~/ossp_practical$ pstree -p $$
bash(360)---pstree(16081)
siddharth@siddharthReddy:~/ossp_practical$
```



```
skill1_process.c
~/ossp_practical

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6
7 int main()
8 {
9     pid_t pid;
10
11     printf("Parent process started\n");
12     printf("Parent PID: %d\n", getpid());
13
14     pid = fork();
15
16     if (pid < 0)
17     {
18         perror("fork failed");
19         return 1;
20     }
21
22     if (pid == 0)
23     {
24         /* Child process */
25         printf("\nChild process created\n");
26         printf("Child PID : %d\n", getpid());
27         printf("Child PPID: %d\n", getppid());
28
29         printf("Child executing 'sleep' using exec()\n");
30         fflush(stdout);
31
32         execlp("sleep", "sleep", "20", NULL);
33
34         /* Only reached if exec() fails */
35         perror("exec failed");
36         exit(1);
37     }
38 }
```

Saving file "~/ossp\_practical/skill1\_process.c"...

C Tab Width: 8 Ln 54, Col 2 INS

# Skill -1 OS

```
29     printf("Child executing 'sleep' using exec()\n");
30     fflush(stdout);
31
32     execlp("sleep", "sleep", "20", NULL);
33
34     /* Only reached if exec() fails */
35     perror("exec failed");
36     exit(1);
37 }
38 else
39 {
40     /* Parent process */
41     printf("\nParent process\n");
42     printf("Parent PID: %d\n", getpid());
43     printf("Child PID : %d\n", pid);
44
45     printf("Parent waiting for child...\n");
46
47     wait(NULL);
48
49     printf("\nChild process completed.\n");
50     printf("Parent process completed.\n");
51 }
52
53 return 0;
54
```

C ▾ Tab Width: 8 ▾ Ln 54, Col 2 INS

```
siddharth@siddharthreddy:~/ossp_practical$ gcc -Wall -Wextra skill1_process.c -o skill1_process
siddharth@siddharthreddy:~/ossp_practical$ ./skill1_process
Parent process started
Parent PID: 16136

Parent process
Parent PID: 16136
Child PID : 16137
Parent waiting for child...

Child process created
Child PID : 16137
Child PPID: 16136
Child executing 'sleep' using exec()

Child process completed.
Parent process completed.
siddharth@siddharthreddy:~/ossp_practical$ strace -f -e trace=process ./skill1_process
execve("./skill1_process", ["/skill1_process"], 0x7ffe88db6678 /* 26 vars */) = 0
Parent process started
Parent PID: 16168
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, strace: Process 16169 attached
, child_tidptr=0x70e36f4a2a10) = 16169

Parent process
Child process created
Parent PID: 16168
Child PID : 16169
Child PID : 16169
```

# Skill -1 OS

```
Child process completed.
Parent process completed.
siddharth@SiddharthReddy:~/ossp_practical$ strace -f -e trace=process ./skill1_process
execve("./skill1_process", ["/skill1_process"], 0x7ffe88db6678 /* 26 vars */) = 0
Parent process started
Parent PID: 16168
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARPID|CLONE_CHILD_SETTID|SIGCHLD, strace: Process 16169 attached
, child_tidptr=0x70e36f4a2a10) = 16169

Parent process
Child process created
Parent PID: 16168
Child PID : 16169
Child PID : 16169
Parent waiting for child...
[pid 16168] wait4(-1, Child PPID: 16168
Child executing 'sleep' using exec()
<unfinished ...>
[pid 16169] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = -1 ENOENT (No such file or
directory)
[pid 16169] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = -1 ENOENT (No such file or d
irectory)
[pid 16169] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = -1 ENOENT (No such file or direct
ory)
[pid 16169] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = 0
[pid 16169] exit_group(0) = ?
[pid 16169] +++ exited with 0 +++
<... wait4 resumed>NULL, 0, NULL) = 16169
```

```
ory)
[pid 16169] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffd77094058 /* 26 vars */) = 0
[pid 16169] exit_group(0) = ?
[pid 16169] +++ exited with 0 +++
<... wait4 resumed>NULL, 0, NULL) = 16169
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=16169, si_uid=1000, si_status=0, si_utime=0, si_stime=0} ---

Child process completed.
Parent process completed.
exit_group(0) = ?
+++ exited with 0 +++
siddharth@SiddharthReddy:~/ossp_practical$ strace -f -e trace=process ./skill1_process 2>&1 | grep -E 'clone|fork|execve
|wait|exit'
execve("./skill1_process", ["/skill1_process"], 0x7fff36e464a8 /* 26 vars */) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARPID|CLONE_CHILD_SETTID|SIGCHLD, strace: Process 16175 attached
[pid 16174] wait4(-1, Parent process started
[pid 16175] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffe0e5c2e18 /* 26 vars */) = -1 ENOENT (No such file or
directory)
[pid 16175] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffe0e5c2e18 /* 26 vars */) = -1 ENOENT (No such file or d
irectory)
[pid 16175] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffe0e5c2e18 /* 26 vars */) = -1 ENOENT (No such file or direct
ory)
[pid 16175] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffe0e5c2e18 /* 26 vars */) = 0
[pid 16175] exit_group(0) = ?
[pid 16175] +++ exited with 0 +++
<... wait4 resumed>NULL, 0, NULL) = 16175
Parent waiting for child...
exit_group(0) = ?
+++ exited with 0 +++
siddharth@SiddharthReddy:~/ossp_practical$ |
```

# Skill -1 OS

```
siddharth@SiddharthReddy:~$ pstree -p
systemd(1)
├─agetty(234)
├─cron(173)
├─dbus-daemon(174)
├─init-systemd(Ub(2))
│   ├──SessionLeader(358)─Relay(360)(359)─bash(360)─skill1_process(16136)─sleep(1613+
│   │   ├──SessionLeader(16138)─Relay(16143)(16139)─bash(16143)─pstree(16156)
│   │   ├──init(7)─{init}(8)
│   │   ├──login(361)─bash(430)
│   │   └─{init-systemd(Ub)}(9)
├─polkitd(11931)
│   ├──{polkitd}(11933)
│   ├──{polkitd}(11934)
│   └─{polkitd}(11935)
├─rsyslogd(12845)
│   ├──{rsyslogd}(12847)
│   ├──{rsyslogd}(12848)
│   └─{rsyslogd}(12849)
├─systemd(405)
│   ├──(sd-pam)(406)
│   ├──at-spi-bus-laun(553)
│   │   ├──dbus-daemon(560)
│   │   │   ├──{at-spi-bus-laun}(554)
│   │   │   ├──{at-spi-bus-laun}(555)
│   │   │   ├──{at-spi-bus-laun}(556)
│   │   │   ├──{at-spi-bus-laun}(558)
│   │   └─at-spi2-registr(588)
│   │       ├──{at-spi2-registr}(589)
│   │       ├──{at-spi2-registr}(590)
│   │       └─{at-spi2-registr}(591)
│   ├──dbus-daemon(549)
│   ├──dconf-service(600)
│   │   ├──{dconf-service}(601)
│   │   ├──{dconf-service}(602)
│   │   └─{dconf-service}(603)
│   ├──xdg-desktop-por(561)
│   │   ├──{xdg-desktop-por}(562)
│   │   ├──{xdg-desktop-por}(563)
│   │   ├──{xdg-desktop-por}(564)
│   │   └─{xdg-desktop-por}(593)
│   ├──xdg-desktop-por(581)
│   │   ├──{xdg-desktop-por}(582)
│   │   ├──{xdg-desktop-por}(583)
│   │   ├──{xdg-desktop-por}(585)
│   │   └─{xdg-desktop-por}(586)
│   ├──xdg-document-po(566)
│   │   ├──fusermount3(577)
│   │   ├──{xdg-document-po}(567)
│   │   ├──{xdg-document-po}(568)
│   │   ├──{xdg-document-po}(569)
│   │   ├──{xdg-document-po}(576)
│   │   ├──{xdg-document-po}(578)
│   │   ├──{xdg-document-po}(579)
│   │   └─{xdg-document-po}(574)
│   └─xdg-permission-(570)
│       ├──{xdg-permission-}(571)
│       ├──{xdg-permission-}(572)
│       └─{xdg-permission-}(574)
├─systemd-journal(2722)
├─systemd-logind(195)
├─systemd-resolve(2982)
├─systemd-timesyn(2785)─{systemd-timesyn}(2786)
├─systemd-udev(2844)
│   ├──{udev-worker}(16140)
│   ├──{udev-worker}(16141)
│   └─{udev-worker}(16142)
├─unattended-upgr(277)─{unattended-upgr}(322)
└─wsl-pro-service(199)
    ├──{wsl-pro-service}(268)
    ├──{wsl-pro-service}(269)
    ├──{wsl-pro-service}(270)
    ├──{wsl-pro-service}(271)
    ├──{wsl-pro-service}(272)
    ├──{wsl-pro-service}(276)
    └─{wsl-pro-service}(307)
```

```
├─xdg-document-po(576)
├─xdg-document-po(578)
├─xdg-document-po(579)
├─xdg-permission-(570)
│   ├──{xdg-permission-}(571)
│   ├──{xdg-permission-}(572)
│   └─{xdg-permission-}(574)
├─systemd-journal(2722)
├─systemd-logind(195)
├─systemd-resolve(2982)
├─systemd-timesyn(2785)─{systemd-timesyn}(2786)
├─systemd-udev(2844)
│   ├──{udev-worker}(16140)
│   ├──{udev-worker}(16141)
│   └─{udev-worker}(16142)
├─unattended-upgr(277)─{unattended-upgr}(322)
└─wsl-pro-service(199)
    ├──{wsl-pro-service}(268)
    ├──{wsl-pro-service}(269)
    ├──{wsl-pro-service}(270)
    ├──{wsl-pro-service}(271)
    ├──{wsl-pro-service}(272)
    ├──{wsl-pro-service}(276)
    └─{wsl-pro-service}(307)

siddharth@SiddharthReddy:~$
```

# Skill -1 OS

```
siddharth@siddharthReddy:~/ossp_practical$ strace -t -e trace=process ./skill1_process 2>&1 | grep -E 'clone|fork|execve|wait|exit'
execve("./skill1_process", ["/skill1_process"], 0x7ffd742757e8 /* 26 vars */) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARPID|CLONE_CHILD_SETTID|SIGCHLD, strace: Process 16195 attached)
[pid 16194] wait4(-1, Parent process started)
[pid 16195] execve("/usr/local/sbin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 16195] execve("/usr/local/bin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 16195] execve("/usr/sbin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = -1 ENOENT (No such file or directory)
[pid 16195] execve("/usr/bin/sleep", ["sleep", "20"], 0x7ffffb6e5fc38 /* 26 vars */) = 0
```